Final Project: Duck Car

ENGS 26 Control Theory

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Introduction

Project Design Goals

The goal of the duck car project was to design an appropriate compensator for this proximity maintenance system. Using an infrared (IR) sensor to detect distance from a target, the car demonstrates the practical application of feedback control in maintaining a fixed position. The duck car should be able to quickly respond to a sudden movement of the wall towards, or away from, the car by quickly accelerating and decelerating to arrive at the prescribed operating distance with minimal oscillation and overshoot. In theory, we should be able to line up many duck cars in a row and they will follow each other as if they are a line of ducks (hence the name "duck" car).

Objectives

We outlined the following, as main objectives for this project:

- The closed-loop system must be stable.
- The steady-state error of the closed-loop system in response to a step input must be zero.
- The transient response to a step command must have a 2% settling time under 0.2-0.3 sec and overdamped (no/very little overshoot).
- The maximum magnitude of the closed-loop frequency response is 1 dB above the DC value.
- The maximum output magnitude of the power amp or any op-amp is 12 V.
- The closed-loop bandwidth (the frequency at which the magnitude of the closed-loop frequency response decreases by 3 dB below the value at DC) must be increased to the greatest possible value.
- The system reduces response to any disturbances and is robust to model uncertainty.
- System must have at least 6 dB gain margin and 30 degrees phase margin.
- The controller may be any type (P, PI, PD, PID, lead, lag, lead-lag, etc.); low-pass or notch-pass filters on a sensor could be considered.

Model

Overview

To understand the behavior of any mechanical system, we start with fundamental laws of motion, $\Sigma F = ma$. Its polar "analogue" is $J\omega = \Sigma\tau$. From there, we can conclude that a change in the angular velocity is proportional to a total of torques applied: in our case, rotor's torque and load torque. The first is proportional to the input voltage and the latter is proportional to some damping coefficient and the angular velocity. From there, we can construct an ODE in the form $\frac{d\omega(t)}{dt} + C_1\omega(t) = C_2u(t)$,

where $u(t)$ is some input. Let's take a Laplace transform of this ODE. We arrive to a simple observation that angular velocity in the s-domain is proportional to a coefficient of gain per input and depends on some time constant: $\frac{\Theta(s)}{U(s)} = \frac{C_2}{s+C_1} = \frac{C_2}{\tau s+1}$. Angular velocity is proportional to linear velocity by a factor of 2π, which leads us to a conclusion that plant's block would have a form of $V(s) = \frac{gain\ per\ input}{\tau s+1}$. Please see the figure with detailed derivations down below.

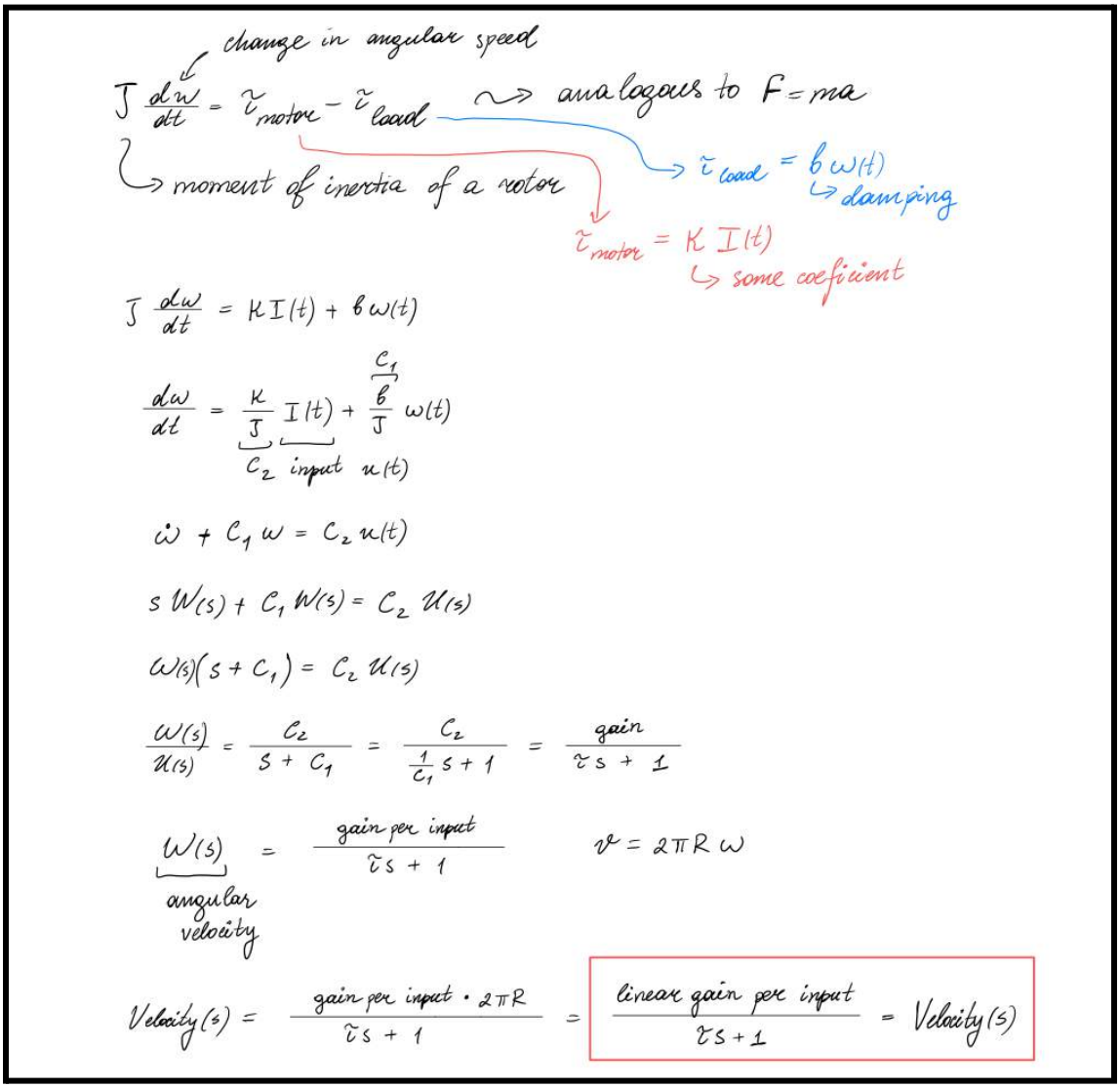


Figure 1: *Derivation of the Plant Block Equation.*

We characterized the closed loop system of the car with three blocks. See Figure 2 below. We are meant to design the compensator and we will model the plant and sensor blocks through experimental results later in this report.

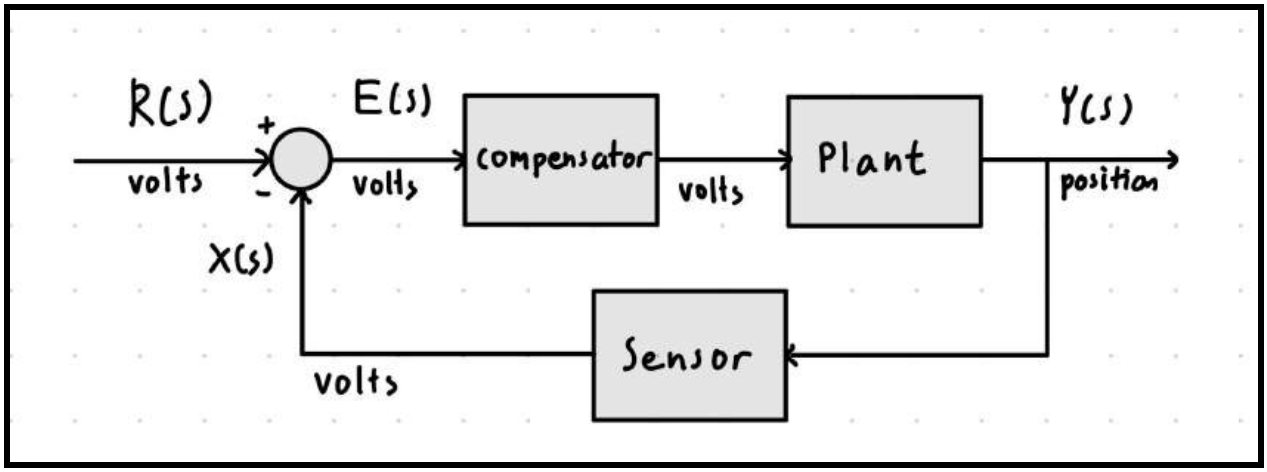


Figure 2: *high-level overview of duck car block diagram.*

Detailed Block Design

See Figure 3 below. The red is meant to show the compensator, which as we will discuss later, is denoted as a PD controller The blue denotes the plant block. For simplicity purposes, we tuned the power amplifier block to unity gain. For the motor and RPM to velocity blocks, we used an equation from a previous lab: $\frac{k}{\tau s+1}$, where k is a proportionality constant of the two blocks and tau is the time constant of the blocks. The mechanical plant output velocity, therefore an infrared sensor that reads distance to the obstacle must have an integrator, which we modeled as $\frac{1}{s}$. For simplicity purposes, we decided to carry on the $\frac{1}{s}$ term into the plant's block. We will discuss how we arrived at sensor gain later in the report.

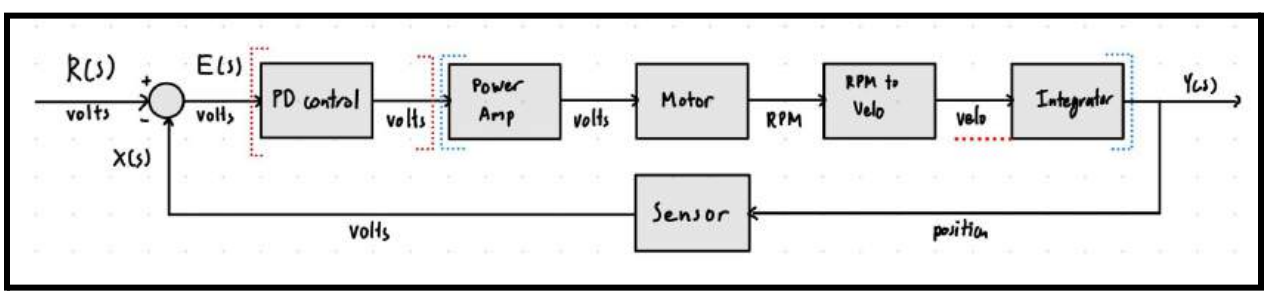


Figure 3: *detailed overview of duck car block diagram.*

Plant Block Characterization

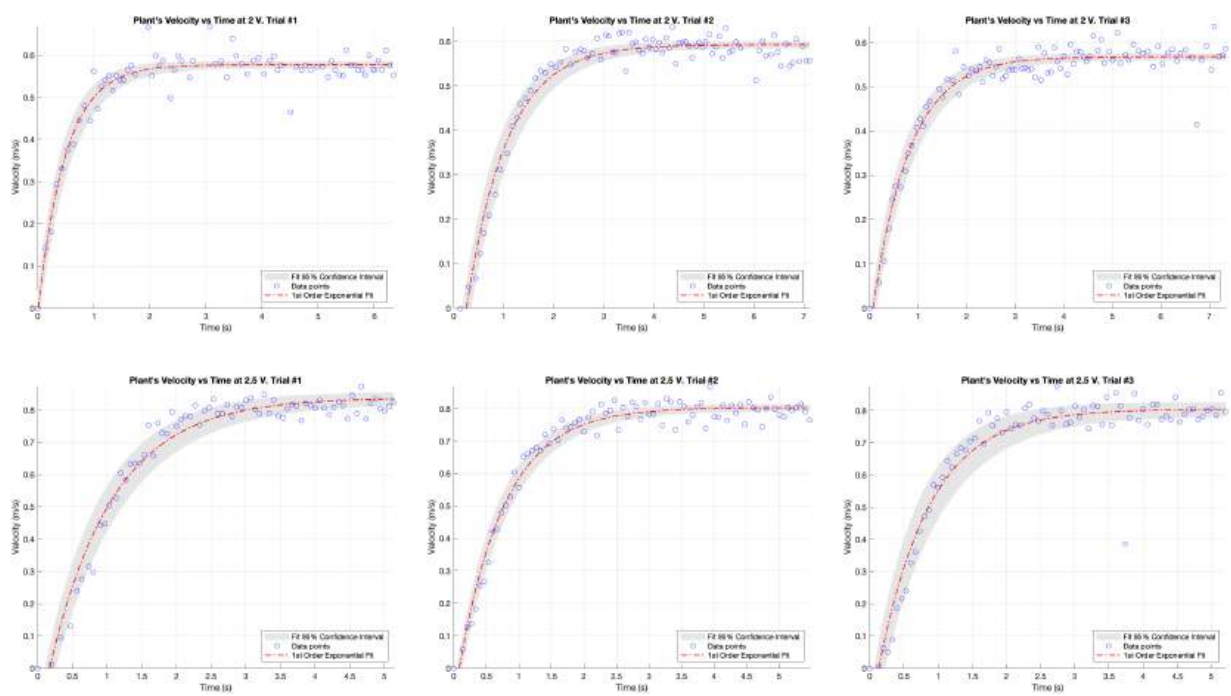
Using our fully derived equations above, we arrive at this equation for the plant block: $\frac{k}{(\tau s+1)}$ combined with an integrator term $\frac{1}{s}$. Let's consider plant as $\frac{k}{(\tau s+1)}$, outputting velocity before

integration term. Let's apply some step input in the form $u(t) = A$, where A is some constant value (unit step function scaled by A). Applying Final Value Theorem, we achieve the following:

$$y(t = \infty) = \lim_{s \to 0} s \cdot \frac{A}{s} \frac{k}{(\tau s+1)} = \lim_{s \to 0} \frac{Ak}{\tau s+1} = Ak$$

Therefore, $k = \frac{y(t=\infty)}{A}$, where A is an input voltage and k is a coefficient of proportionality. This shows that we can find the gain by evaluating a steady state response.

To find the rate of change, K value and then a time constant ($\frac{1}{K}$), we created an open-loop system with which we ran a DC voltage and let the car run to characterize its first-order velocity behavior. In doing so, we took the steady state (SS) velocity divided by our reference voltage and averaged the values we calculated to find the rate of change K (see Figure 3 below). Furthermore, from the velocity function we were able to derive a first-order system that led us to find a steady-state velocity and time constant (τ) from the function of the form: $A(1 - e^{-Kx}) = y(t)$. By using the equation $\frac{1}{K} \triangleq \tau$ we were able to find the time constant.



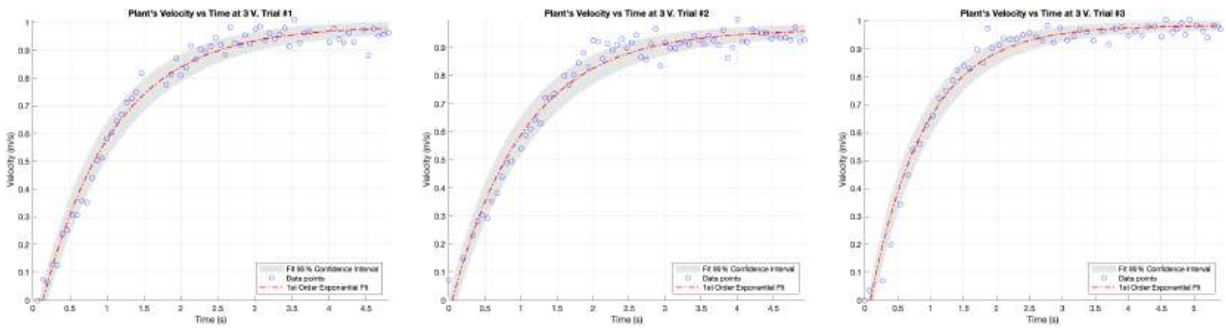


Figure 4: *From left to right: trials 1–3; from top to bottom: reference voltages 2, 2.5, and 3 V.*

Ref. Voltage	Steady State Velocity, m/s		Gain, K $\frac{S.S.\,Voltage}{Ref.\,Voltage}$	Time constant τ, $\frac{1}{sec}$	
	Mean	95% CI		Mean	95% CI
2 V	0.579	(0.571, 0.587)	0.2895	0.685	(0.588, 0.737)
2.5 V	0.816	(0.799, 0.832)	0.3264	0.725	(0.654, 0.811)
3 V	0.979	(0.960, 0.997)	0.326	0.83[1]	(0.764, 0.911)
Average	**N/A**	**N/A**	**0.324**	**0.746**	**(0.6, 0.8)**

Figure 5: *detailed overview of duck car block diagram.*

After characterizing the car's open-loop behavior, we encountered sensor malfunctions, possibly due to a loose connection in the wiring, accidental circuit shorting, worn off components, or other factors. Due to the intricate design of the machine, replacing the sensor alone was not feasible, so a new car had to be used. Since the new car employs the same motor and sensor, we assumed the time constant and sensor gain would fall within the same range as the previously determined values. By adopting the most conservative estimates, we retained the flexibility to later adjust values of the circuitry components to optimize its operation. By examining the 95% confidence intervals of the time constants from a previous car, we decided to adopt the most conservative estimate, $\tau = 0.6\,\frac{1}{sec}$. Starting with the lowest time constant estimate would lead to our system responding too fast rather than too slow. In the worst-case scenario, the designed compensator would result in a car operating at a slower speed, which is preferable to having a car that moves too fast and risks crashing into obstacles. If the car does end up being too slow, we plan to increase the gain of the compensator to optimize the

[1] Due to the camera movements, the time constant estimate for the Voltage Reference of 3 V was based on the trial No. 3 alone, while the steady state estimate was based on all three trials.

response time. The gain of the plant is proportional to the input voltage and the steady state response of the system, yielding a mean value of $K = 0.324$.

From the values derived above, we can fill in our plant block equation as: $\frac{0.324}{s(0.6s+1)}$. On its own, the motor has been found to produce a stable response in its operation alone, achieving a certain steady state velocity after a certain amount of time. Therefore, it's important not to introduce any instability in future design choices.

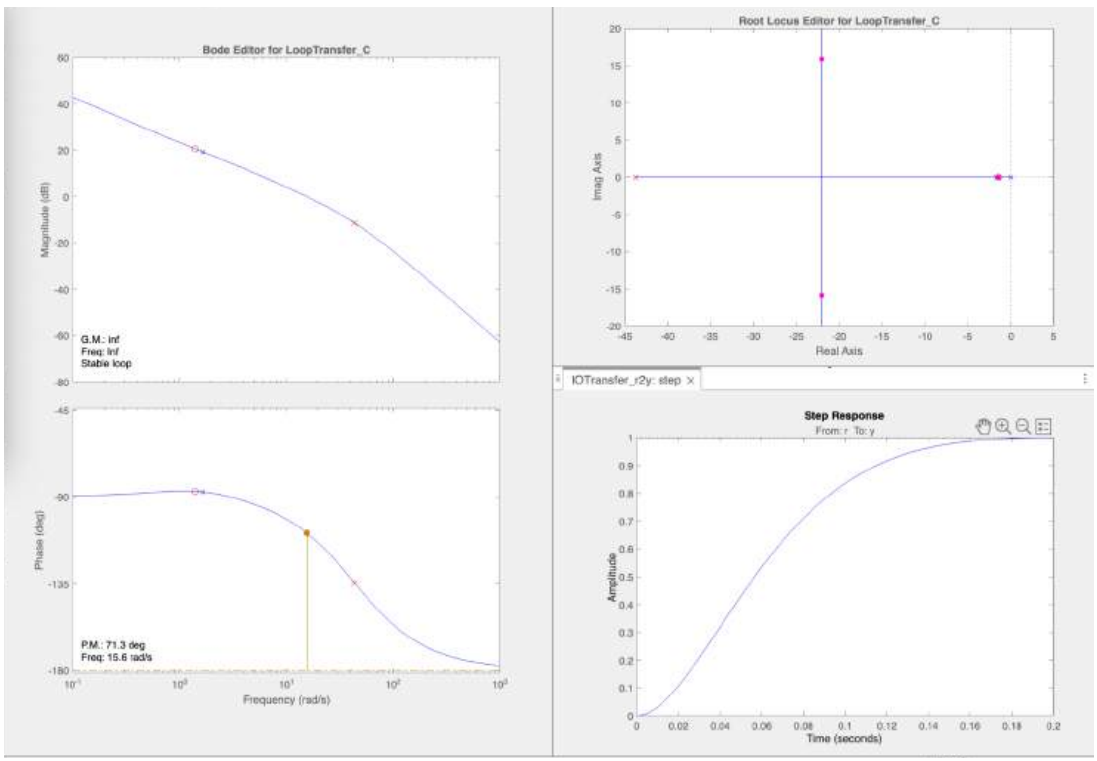


Figure 6: *Motor's Operation Response*

Sensor Block Characterization

To characterize the sensor gain we measured the sensor voltage output at distances from a wall (0-30 inches). From this data, we linearized around an operating point (we chose ~4 inches or ~10 cm) to find the sensor gain (in V/m units). From our data, we were able to characterize the gain of the sensor about the chosen operating point as roughly -8.94 [V/m]. This value aligns with the acceptable range which Bob had specified as roughly eight to ten for this gain (see Figure 7 below for further details).

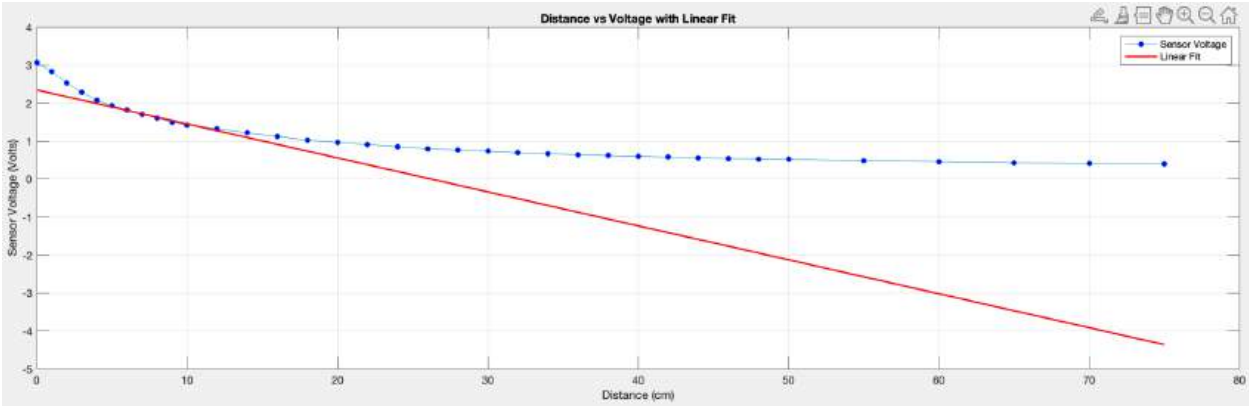


Figure 7: *Sensor block diagram measurements with linearized line of best fit around operating point at ~10 cm:* $y(t) = -8.94x + 2.3443$ *where x is in meters and y is in volts.*

Uncompensated System Analysis

Using MATLAB and the sisotools function, we were able to determine the stability of our plant + sensor blocks. In doing so, we were able to determine that we had two goals/specifications in mind when designing our compensator. First, we must decrease the settling time of our full closed-loop system given that the transient response is quite slow. Second, we want to get rid of our oscillation. If the zero introduced by the PD controller is too close to the pole of the system, it can create overshoot or oscillations due to pole-zero cancellation effects. Thus, we must make sure our zeros and poles are separated by at least an order of magnitude. We must implement these specifications without compromising the stability of the system (all roots already in the RHP). See Figure 8 below.

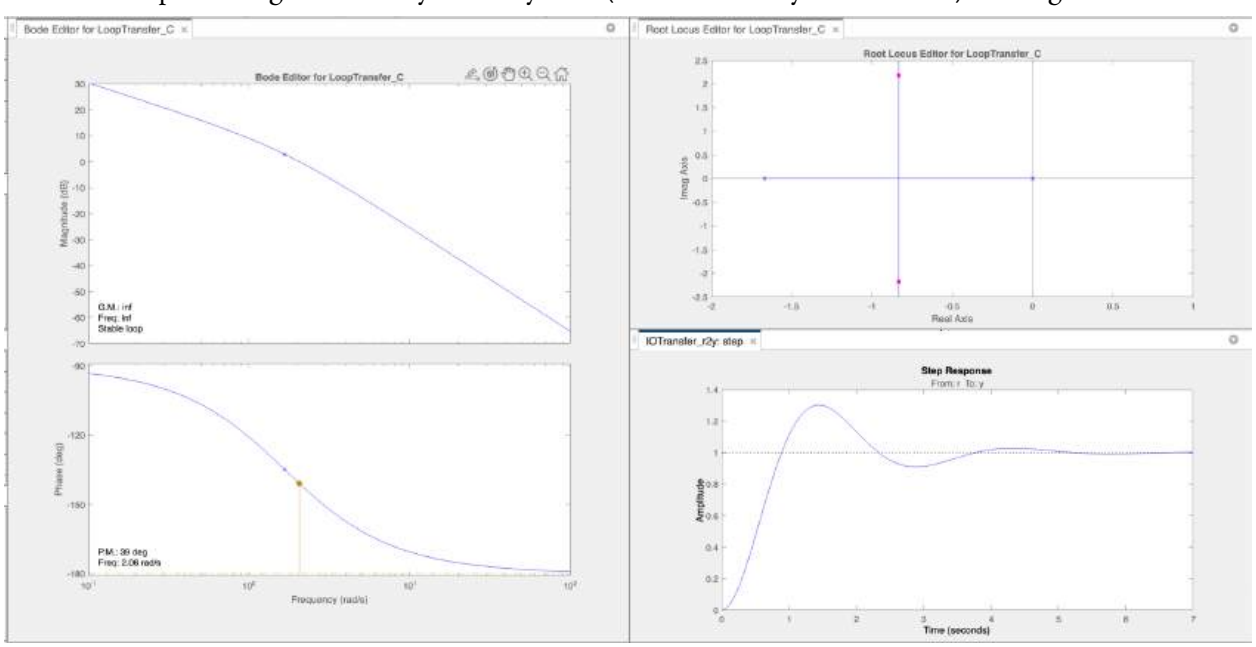


Figure 8: *SISOtool results for open-loop system (ie. no compensator).*

Controller Design & Simulation

Controller Type & Analog Circuit Components

One of the primary design issues was choosing a controller type for our compensator. We were most familiar with the PID and PD controllers from the lab and class. We carefully considered our options and landed on using a PD controller. In general, PD controller's improve damping and reduce oscillations, both of which we want for our system. The derivative term counteracts oscillations, stabilizing the system. Additionally, adding the integral term can reduce damping, potentially increasing overshoot and introducing oscillations. The integral action amplifies slow, persistent errors, which may cause the system to overreact and oscillate if the integral gain is not tuned correctly. We found that the PD controller would also yield a faster transient response due to the absence of integral action. Also, since steady-state error is not a major concern in our application, the integral component of a PID controller is unnecessary. Additionally, the simplicity of a PD controller makes it easier to design, tune, and implement, ensuring a streamlined and efficient control solution. One of the concerns we had about PD compensators was the fact that differentiation could introduce noise. We proceeded, keeping that in mind in case we would need to consider other compensator choices (however, it wasn't the issue as we found later).

In the magnetic levitation lab ("Lab 5") we learned an effective design for a PD controller (see Figure 9 below) with a stop resistor.

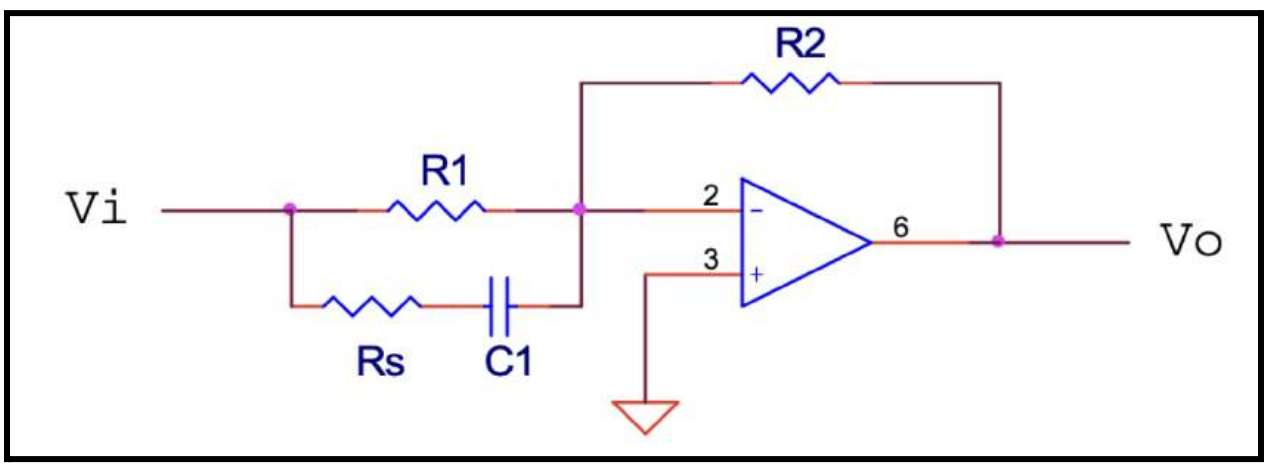


Figure 9: *PD controller design with stop resistor from Appendix B in Lab 5.*

Analog Circuit Components & Compensated System Design

We ran some further sisotools analysis in MATLAB to design an appropriate compensator (with one real zero and one real pole). Note, that we attempted to keep the pole and zero at least an order of magnitude separated from one another to avoid the proximity effect discussed above. Additionally, we knew that a higher gain would yield a snappier response with less oscillation, so we tried for a gain of ~ 5, but given the parts available to us in the lab, we found that ~ 4.25 was sufficient

(we will further discuss this point in the testing section). See Figure 10 below for MATLAB sisotools behavior and Figure 8 for resistor and capacitor value derivations. Additionally, when modeling a compensator, we paid attention to not creating an accidental resonance peak at high frequencies, which leads to a variety of issues such as "op amp saturation, actuator's overheating, etc." as noted by Appendix C of Lab 5.

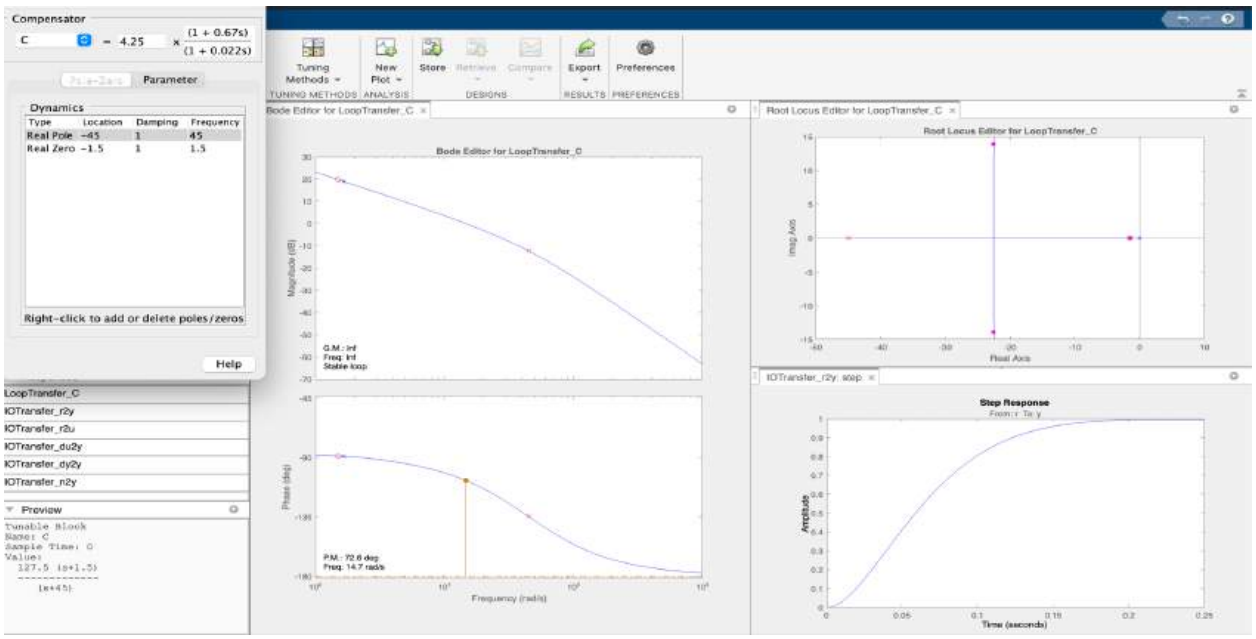


Figure 10: *SISOtool results for closed-loop system with compensator transfer function.*

By implementing this design, we get a stable response with a 72.6 deg phase shift, adhering to a listed specification of at least 30 deg. Looking at a Bode plot, we arrived having an infinite gain margin. By looking at the Root Locus plot, we ensured that there are no roots in the right half-plane.

After characterizing our main parameters, we arrived at the following system (see figure below).

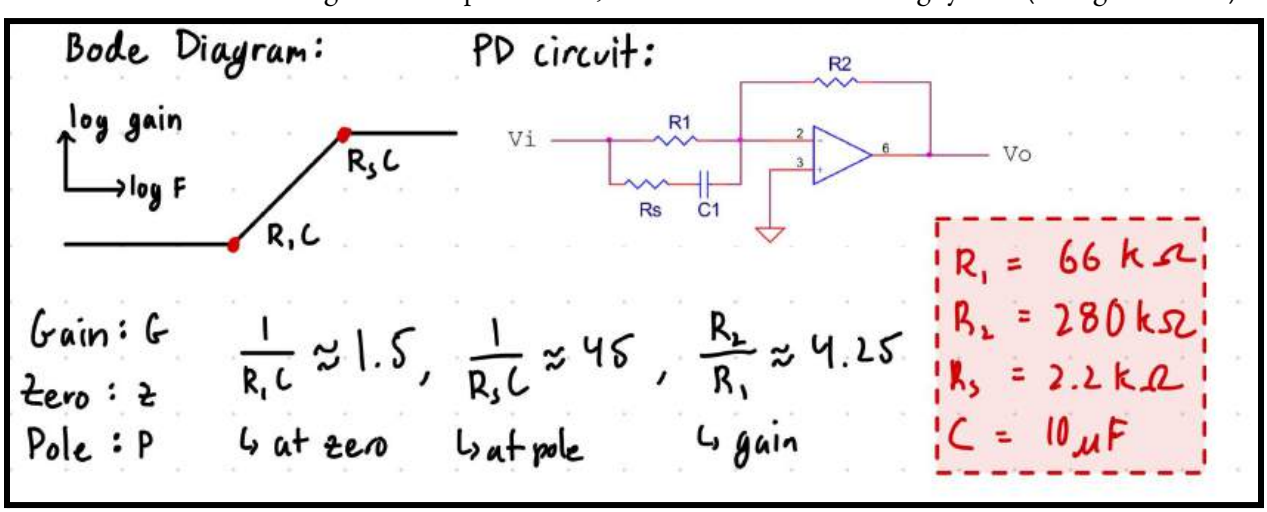


Figure 11: *Compensator resistor and capacitor value derivations.*

This system could be represented with the following block diagram (see figure below).

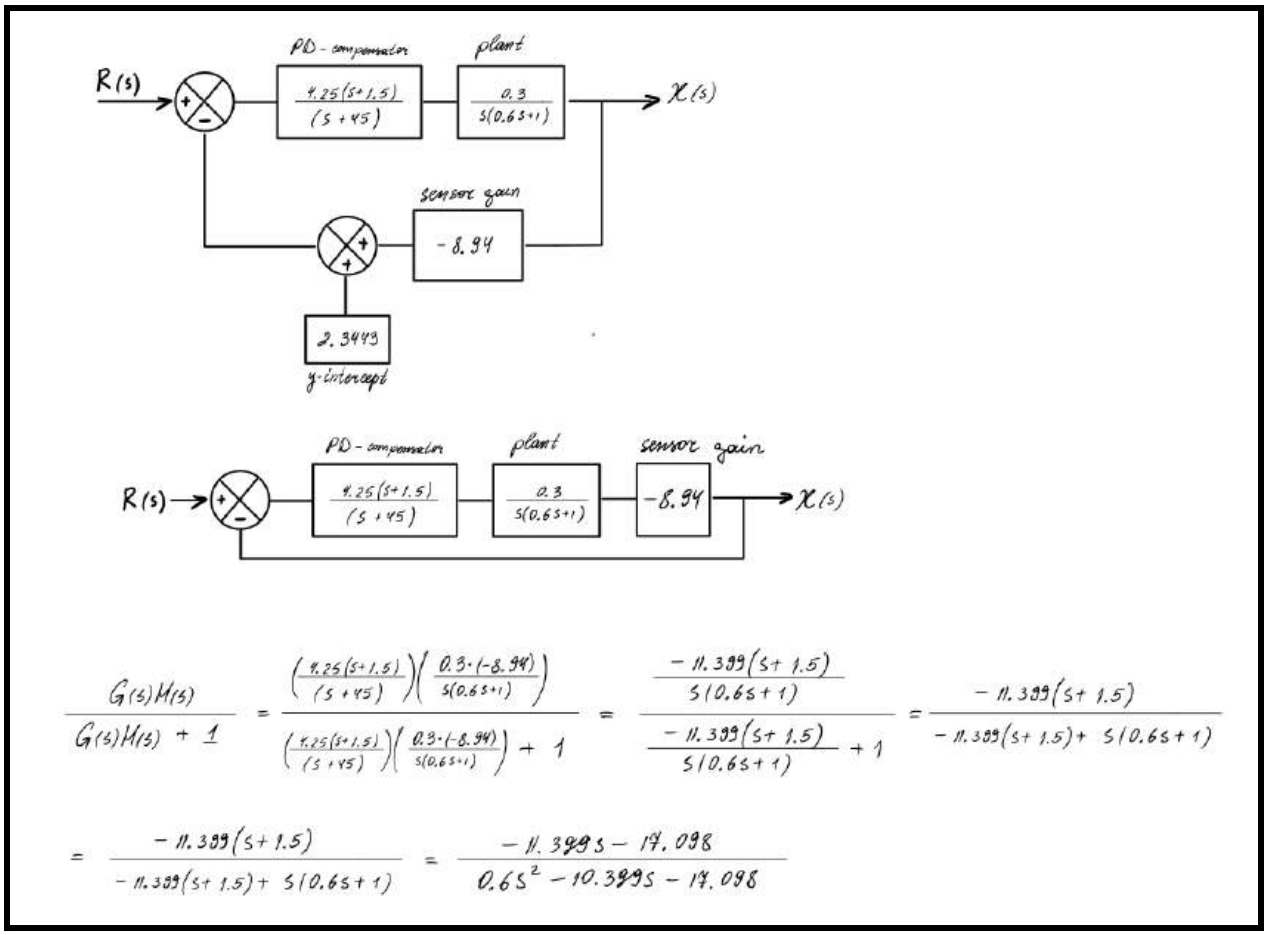


Figure 12: *Block Diagram of a Characterized System*

Testing

Experimental Results

We initially tested our circuit with a pole of -15, zero at -1.5, and gain of 5 (see Figure 14 below). This left us with some oscillatory behavior and a longer settling time than we wanted (~0.4 seconds). We also tested the same system with a gain of 1 and pole at -45, we again found a settling time that was too long (see Figure 13 below), so we combined the two to settle on the final numbers (see Figure 10 above).

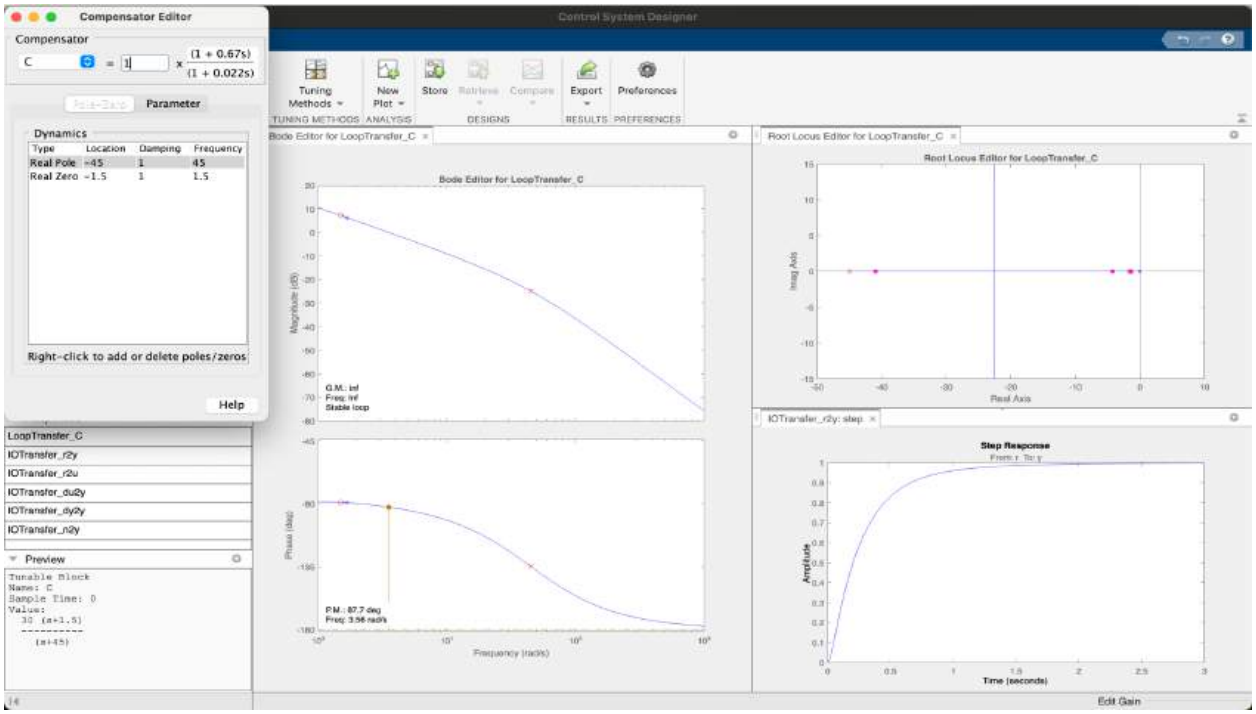


Figure 13: *SISOtool results for closed-loop system with compensator transfer function (using gain of 1 and final pole and zero values).*

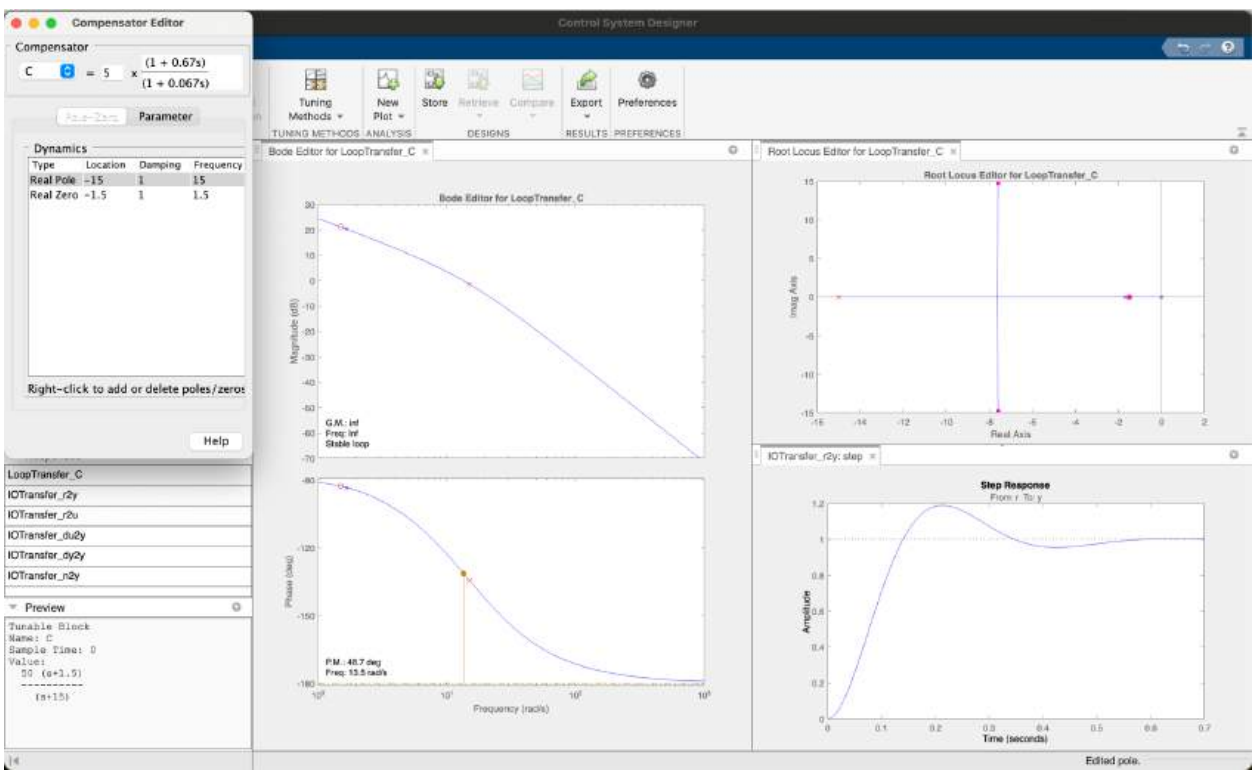


Figure 14: *SISOtool results for closed-loop system with compensator transfer function (using gain of 5 and pole of 15, note the zero value is the same as our final transfer function).*

Experimental Video Link

Please use the link to access our video demonstration: **https://youtu.be/sOCTxiUCPxs**

Final Schematics

Real resistances of R_1, R_2, R_s and capacitance C were measured to complete the final design schematics (see figures 15 and 16 below).

	Modeled	Executed
R_1	66 kΩ	68.19 kΩ
R_2	280 kΩ	220.75 kΩ + 68.2 kΩ
R_s	2. 2 kΩ	2.198 kΩ
C	10 μF	10.4 μF

	Modeled	Executed
Pole	43.71	43.75
Zero	1.46	1.41
Gain	4.242	4.237

Figures 15 (left) and 16 (right): *Figure X demonstrates our resistor and capacitor values modeled in theory, and our final values measured in practice. Figure Y demonstrates the same as figure X, but for the pole, zero, and gain of the compensator.*

Given these final measured resistor and capacitor values, we modeled the final poles, zeros, and gain in sisotools using the mathematical derivations shown in Figure 11.

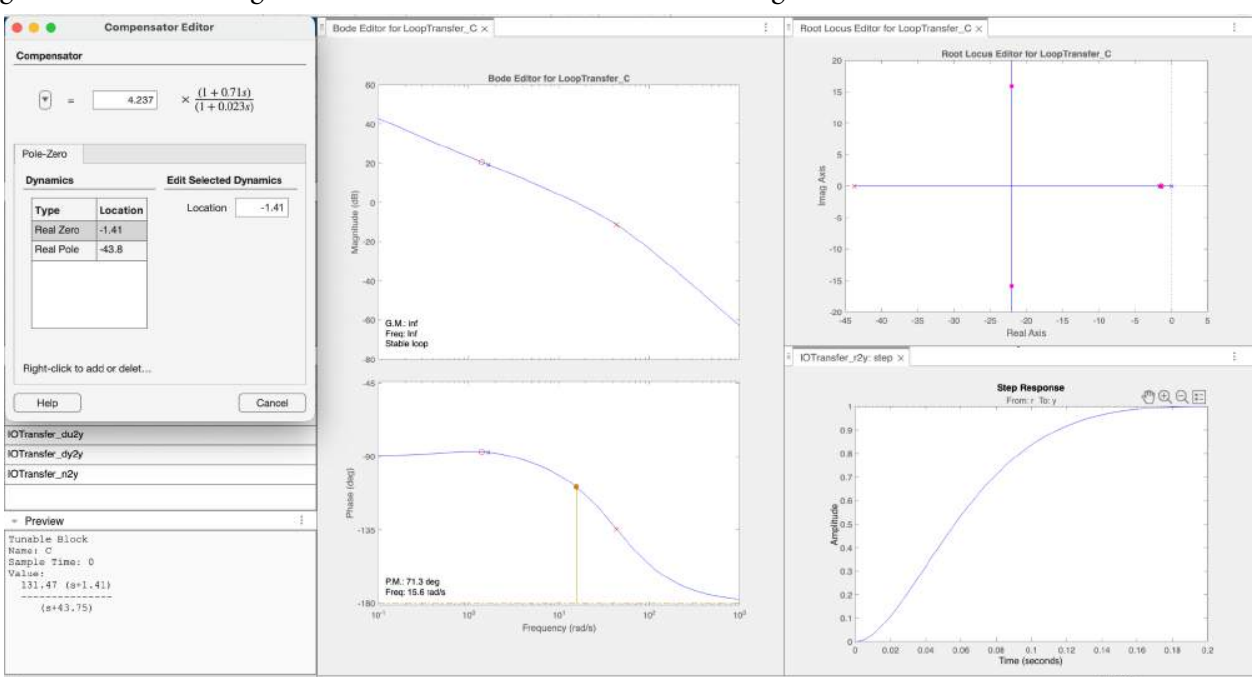


Figure 17: *Response of the System of the Final Executed Design.*

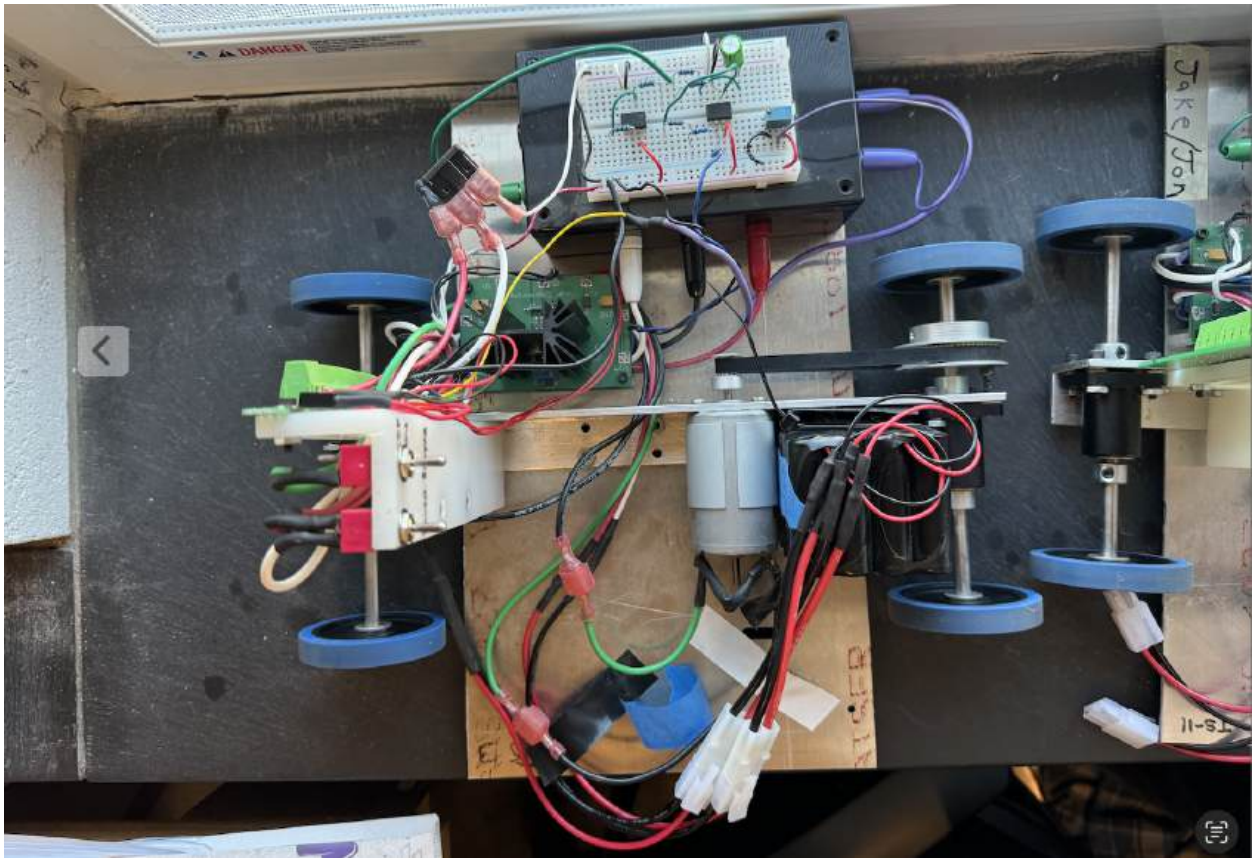


Figure 18: *Final circuit implementation. Note that we have an inverted unity gain circuit on the far left of the breadboard which we did not end up implementing.*

Evaluation

Actual vs. Expected Results

Our actual results were quite close to our expected results. This is notable given the fact that we ran into many troubles with our car. We initially were given car 8. We ran into many problems including blown fuses, bent axles, and most notably, our sensor fried in the end. Thus, the day of our demonstration, Bob gave us a new car (car 13). Given that we how calculated a range for our time constant (τ), we used 0.6 to model our plant transfer function and implemented our circuit in the new car.

In the end, our sisotool's compensator design behavior was modeled quite accurately. We did note that the simulation modeled slightly less oscillation than the actual results. In practice, our car would oscillate before coming to a stop very slightly if we quickly moved our foam wall away from the sensor (as can be seen in the experimental results section video).

The big difference we noticed between our actual and simulated results was in our sensor gain and operating distance characterization. We again must note that given our last-minute switch from car

8 to car 13, Bob told us not to re-measure sensor voltages. Thus, our sensor block characterization would have been slightly different than the actual car.

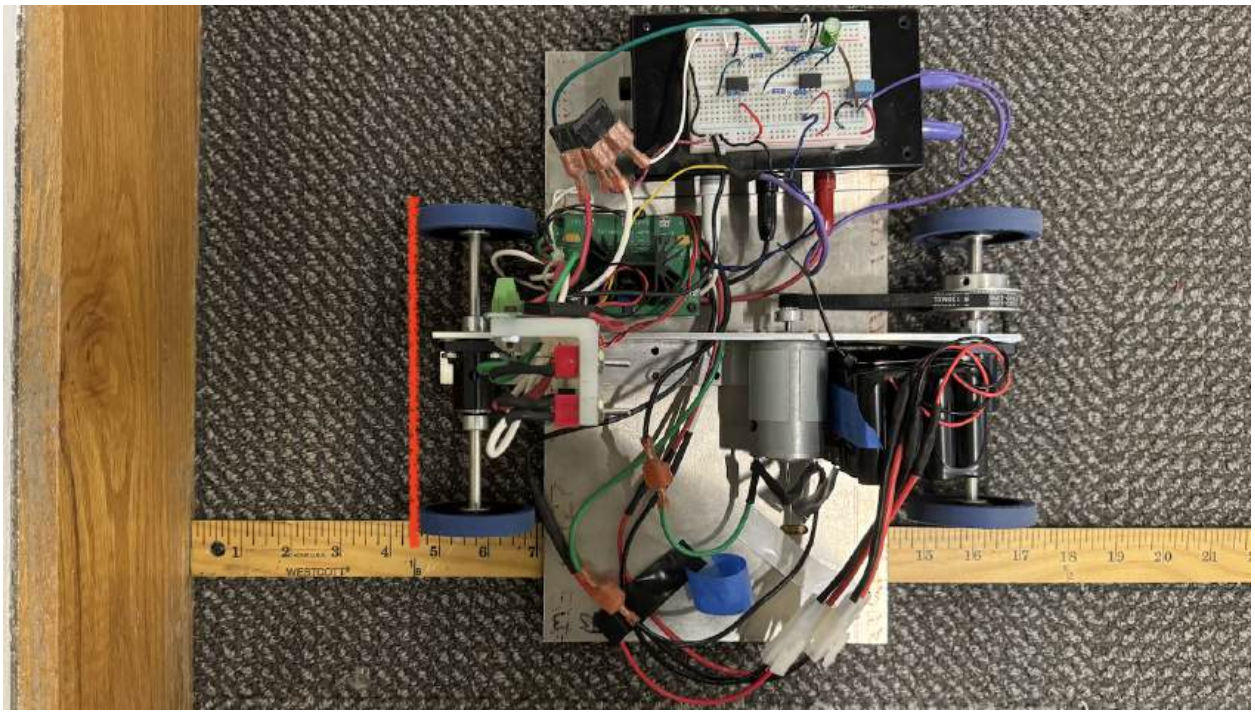


Figure 19: *Operating point measurement (~4.5 inches or ~11.4 centimeters).*

As we can see in Figure 16, the operating point was around 11.4 centimeters. Given our sensor characterization (linearization) (see Figure 7) of $y(t) =- 8.94x + 2.3443$ (x in meters, y is voltage), we had hoped that setting our potentiometer to ~1.4V would yield an operating point of ~10.5 cm. This is almost a full centimeter off. We can attribute this to a slight mischaracterization of the sensor block since we were trying to linearize a fundamentally non-linear system.

Conclusion

Lessons Learned

The Duck Car project provided valuable hands-on experience in implementing control theory concepts in a real-world application. The decision to use a PD controller over a PID controller taught us important lessons about controller selection - specifically that simpler solutions can often be more effective when they match the system requirements. In this case, the PD controller's ability to reduce overshoot and improve response time was sufficient for our position control needs, without the added complexity of an integral term. Perhaps the most valuable part of any engineering is the experience of debugging the hardware. We blew both our fuses out and had to replace them and actually ended up blowing our sensor out by accidentally wiring the voltage rails into the sensor when collecting data to characterize it. Overall, this project and hurdles we had to overcome, very much tested our

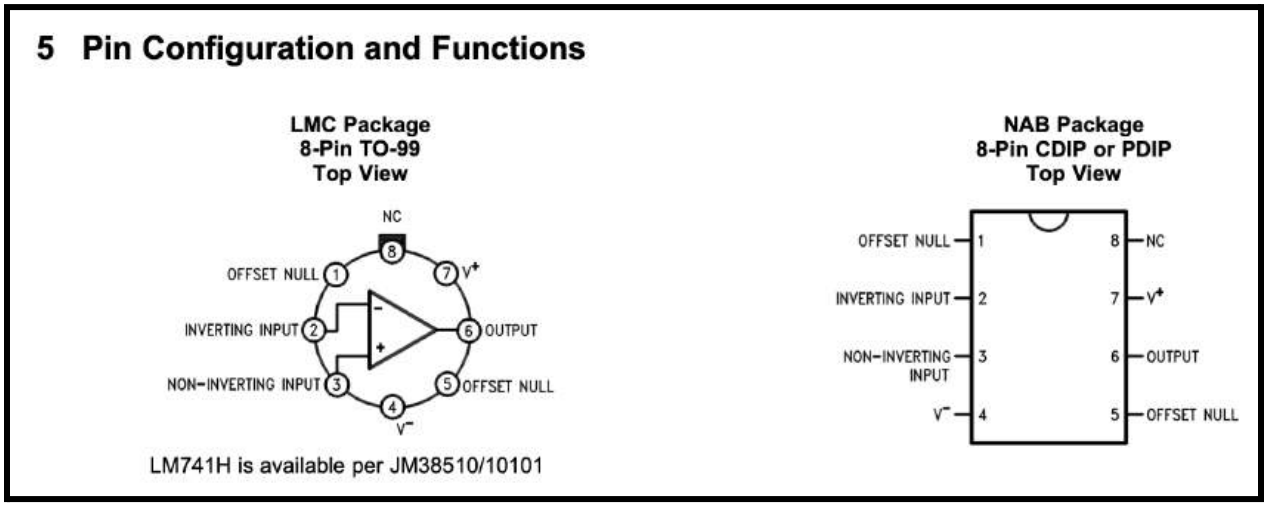
understanding of the system and all of its parts as we tried to understand what and why things were going wrong.

Final Thoughts

In order to navigate this project, it's been found crucial to understand every concept learned in control theory and have intuitive insight into the math to quickly, effectively, and efficiently complete this project. Most importantly, consult Bob to understand the intricacies of the system. For example, we did not realize our power amplifier was not precisely tuned and thus we had to re-do our plant assumptions last minute. Understanding the ins and outs of the circuit and system will make the compensator design and implementation much smoother.

Appendix

Appendix A: Used in the PD controller (compensator). This diagram illustrates the pin configurations for the LM741 operational amplifier in two package types: the LMC (8-pin TO-99) and NAB (8-pin CDIP or PDIP). It identifies the locations of key pins, including the inverting and non-inverting inputs, output, power supply pins (V^+ and V^-), and offset null connections. The LM741 is a general-purpose op-amp commonly used for amplification, buffering, and analog computations in circuits.



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